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Continuous Assessment:

Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines

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Assignment submitted in partial fulfillment of the requirements for the Applied Machine Learning in course in the Data Analytics program at City College Dublin – Dublin Campus.

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Continuous Assessment: Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines

Course: Applied Machine Learning

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Declaration:

I hereby declare that this assignment is my own work and has not been submitted, either in whole or in part, for any other academic purpose. All sources of information have been appropriately acknowledged.

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1. Modelling section and GridSearchCV

Given its efficiency and ability to generate results with easy and quick interpretation⁵, logistic regression was the methodology chosen for the study. Using hyperparameters in the scikit-learn pattern, considering the baseline model ($C=1.0$, penalty L2, solver lbfgs), achieved the following ROC-AUC validation scores: 0.8352 and 0.8631 for H1N1 vaccine prediction and seasonal vaccination, respectively. GridSearchCV with stratified cross-validation in 5 folds systematically analyzed the regularization strength ($C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$), penalties (L1, L2), and optimization algorithms (library saga), resulting in optimal configurations of $C=10$, penalty L2, solver saga for H1N1 (ROC-AUC: 0.8398, $\Delta+0.0046$) and $C=1$, penalty L2, solver linearity for seasonal vaccine (ROC-AUC: 0.8639, $\Delta+0.0008$). Although statistically measurable, the advances in performance remain modest, indicating that the standard hyperparameters achieve optimal configurations for this dataset and work.

This parameter has three limitations. First, the assumption of linearity restricts the ability to model complex interactions between variables observed in behavioral vaccination datasets, where synergistic effects between demographic, attitudinal, and access variables likely exist³. Secondly, even with sample stratification, the notorious class imbalance for H1N1 (78.8% negative class) indicates prediction bias among the majority classes, as seen by the asymmetry between precision and recall, where recall (0.4821) showed significantly lower performance than precision (0.653), which may point to a systematic under-identification of vaccinated individuals⁴. Lastly, sensitivity to variable scaling and multicollinearity was observed, requiring pre-processing with StandardScale and increasing the standard errors of the coefficients where correlated predictors exist⁷. In comparison to the study by Wang⁹, which had a ROC-AUC of 0.87 using the same resource with clinical variables, indicates that the engineering strategy of several variables may, inappropriately, capture predictive signals, of non-linear transformation, to increase the expressiveness of the model within the logistic framework.

2. Evaluation, Conclusions

Regarding vaccination forecasting, impact assessments identify asymmetrical costs associated with classification errors. False positives imply resource allocation, such as vaccine doses and prepared communication materials, which tend to result in sunk costs of less than €10 per person in campaigns. False negatives, on the other hand, represent missed intervention opportunities with cumulative social costs, including preventable disease transmission, treatment expenses, and productivity losses, exceeding €1000 due to influenza⁸. The instability in the accuracy-recall of the current model (H1N1: 0.6534 vs 0.4821) points to systematic under-identification in positive cases, demonstrating that recalibrating the limits towards optimizing sensitivity would maximize practicality in public health, even with increased false positive rates.

Recommendations for implementation include, integration by REST API with electronic health record systems, enabling real-time risk stratification along the clinical encounters, concurrent with weekly model retraining protocols to act on the temporal concept of vaccination attitudes². Among the critical limitations are the linear constraints of Logistic Regression that make it impossible to model interaction effects, the archaic dataset (2009 survey data), and the considerable class imbalance of H1N1.

Future priorities include ensemble methods (Random Forest, XGBoost) that demonstrate best performance in behavioral prediction tasks¹, deep learning architectures that capture non-linear attribute interactions, and real-time learning frameworks for dynamic model updates⁶. Advanced attribute engineering with polynomial terms, domain-specific interaction variables (age×comorbidity, attitude×access), and vaccination history estimates performance gains approaching benchmark of 0.87 ROC-AUC according Wang⁹.

3. Reference

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The exponential numbers in the words represent their respective references. These are enumerated as seen.